

CAGLIARI ELMAS, Italy

WMO: 165600

Lat: 39.244N

Lon: 9.060E

Elev: 2

StdP: 101.30

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.2	2.9	-2.6	3.0	5.8	-0.9	3.5	6.1	14.7	14.3	12.5	14.0	3.0	320	0.480

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.7	34.2	21.8	32.8	21.7	31.1	21.6	25.8	29.3	24.9	28.5	24.1	27.8	5.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.0	28.0	23.9	18.8	27.4	23.0	17.7	26.6	79.6	29.8	75.6	28.6	72.3	28.0	31.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.0	9.7	8.6	-0.9	37.8	1.3	1.6	-1.8	39.0	-2.6	40.0	-3.3	40.9	-4.2	42.1		
			-1.7	27.1	1.3	1.5	-2.6	28.1	-3.4	29.0	-4.1	29.8	-5.0	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	10.2	10.3	12.4	15.0	18.8	23.1	26.0	26.1	22.7	19.5	14.7	11.4
	DBStd	6.19	2.48	2.51	2.44	2.19	2.42	2.59	2.06	2.03	2.31	2.53	2.79	2.63
	HDD10.0	76	27	24	7	1	0	0	0	0	0	0	2	15
	HDD18.3	1127	252	224	183	101	24	1	0	0	1	17	112	214
	CDD10.0	2838	34	33	83	151	272	392	494	499	382	294	144	60
	CDD18.3	846	0	0	0	1	37	142	236	241	132	53	3	0
	CDH23.3	7170	0	0	1	9	199	1144	2332	2405	885	193	3	0
	CDH26.7	2550	0	0	0	0	29	357	959	993	196	16	0	0
Wind	WSAvg	4.1	3.7	4.0	4.3	4.4	4.5	4.5	4.7	4.2	3.8	3.6	3.5	3.5
Precipitation	PrecAvg	422	45	50	40	37	23	11	3	7	32	54	61	56
	PrecMax	630	138	169	163	119	106	69	28	35	98	167	182	159
	PrecMin	240	2	2	0	3	0	0	0	0	0	4	1	7
	PrecStd	106	34	37	32	24	22	14	6	10	28	40	43	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.2	19.1	22.0	24.2	29.8	34.9	36.6	36.9	32.2	28.2	23.7	19.2
		MCWB	14.5	13.4	14.4	15.0	18.3	21.1	21.5	20.7	21.3	20.9	18.0	15.6
	2%	DB	16.8	17.3	20.0	22.4	27.1	32.0	34.2	34.2	30.1	26.6	21.9	18.1
		MCWB	13.6	12.9	13.5	15.1	18.0	20.9	21.2	22.4	21.1	20.5	17.5	15.0
	5%	DB	15.8	16.2	18.7	21.1	25.2	30.1	32.3	32.8	28.9	25.2	20.9	17.1
		MCWB	12.8	12.4	13.2	14.7	17.4	20.2	21.4	22.3	21.1	19.9	17.2	14.4
	10%	DB	14.9	15.1	17.2	19.9	24.0	28.8	31.1	31.1	27.8	24.1	19.9	16.1
		MCWB	12.0	11.7	12.9	14.4	17.0	20.0	21.4	22.4	21.0	19.5	16.6	13.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.4	15.0	16.1	17.6	21.0	25.3	26.7	27.7	25.5	24.1	19.7	16.8
		MCDB	17.2	17.2	19.4	21.0	24.9	29.5	30.5	30.8	27.6	25.7	21.4	18.3
	2%	WB	14.2	13.8	15.0	16.5	19.7	23.5	25.4	26.2	24.2	22.5	18.7	15.7
		MCDB	16.1	16.3	18.0	19.9	24.0	27.7	29.1	29.7	27.0	24.7	20.7	17.5
	5%	WB	13.2	13.0	14.2	15.8	18.9	22.4	24.4	25.4	23.4	21.4	17.8	14.8
		MCDB	15.1	15.5	17.0	19.3	23.1	26.9	28.3	29.0	26.5	23.8	20.1	16.8
	10%	WB	12.4	12.2	13.6	15.2	18.2	21.5	23.6	24.5	22.4	20.5	16.9	13.9
		MCDB	14.4	14.8	16.2	18.5	22.3	26.1	27.8	28.3	26.0	23.1	19.2	15.8
Mean Daily Temperature Range	5% DB	MDBR	8.5	9.0	9.4	9.5	9.9	10.5	10.9	10.7	9.8	9.1	8.3	8.2
		MCDBR	9.0	9.9	11.1	11.3	11.8	12.2	12.4	12.5	10.6	9.6	9.1	8.6
		MCWBR	6.5	6.5	6.4	5.9	5.4	5.2	5.2	5.9	5.1	5.3	5.9	6.2
	5% WB	MCDBR	8.1	9.1	9.3	9.6	10.2	10.4	10.5	10.6	9.0	7.9	8.0	7.8
		MCWBR	6.5	6.6	6.3	5.9	5.6	5.8	6.1	6.5	5.7	5.2	6.0	6.2
Clear-Sky Solar Irradiance	taub	0.345	0.365	0.405	0.444	0.459	0.433	0.448	0.438	0.435	0.410	0.382	0.353	
	taud	2.442	2.373	2.257	2.161	2.157	2.277	2.241	2.278	2.279	2.317	2.351	2.409	
	Ebn at Noon	818	850	848	834	828	848	832	831	808	786	761	775	
	Edh at Noon	83	101	125	146	150	133	137	129	121	106	90	80	
All-Sky Solar Radiation	RadAvg	2.10	2.98	4.26	5.48	6.56	7.54	7.62	6.70	5.02	3.61	2.31	1.89	
	RadStd	0.19	0.26	0.43	0.39	0.46	0.26	0.24	0.31	0.19	0.25	0.16	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	-0.69	-0.91	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.29	N/A	N/A	N/A	-0.50	-0.61	N/A	N/A	N/A	N/A

Nomenclature: See separate page